

The DIP version is convenient for prototyping, while the SMD version is ideal for the final build due to its extremely small size. Despite its compact footprint, the ATtiny85 provides sufficient program memory, processing capability, and enough input/output (I/O) pins to interface with all essential components of this mini Game Boy system.

For display, the SSD1315 OLED module, measuring approximately 1.5cm, is used. Its small size is key

to achieving the ultra-compact form factor. The display interfaces with the ATtiny85 via the I²C protocol, using pins PB0 and PB2, ensuring minimal wiring and efficient communication.

User input is implemented using a single potentiometer. This simple control is sufficient for basic gameplay functions such as movement and speed control. Games like Pong, Flappy Bird, and other minimal arcade-style applications can be easily developed. Since the system is fully programmable, custom games can also be created and uploaded.

Power management is handled by a 1N5817 Schottky diode, enabling efficient and safe supercapacitor charging. The circuit can be powered either from a 2.7V DC source or directly through a small solar cell connected via pads PAD1 and PAD2, enabling completely battery-free operation.

The supercapacitor (2F, 2.7V) acts as the primary energy storage element. It charges within seconds and can power the device for several minutes. In addition to fast charging, it offers a significantly longer

charge-discharge cycle life than conventional batteries, making the design more durable and sustainable.

Software

The Arduino platform used in the system makes it easy to create and program games for this tiny Game Boy. One of the most exciting aspects is that not only is the hardware custom-built, but the games themselves can also be fully designed and personalised according to individual creativity.

Before writing or uploading any game code, a few essential setup steps are required. First, install ATtiny board support in the Arduino IDE so that it can recognise and program the ATtiny85 microcontroller. This can be done by adding the appropriate Board Manager URL and installing the ATtiny core.

Next, install the SSD1315 OLED library. This library is necessary for controlling the display and rendering graphics, enabling games to run smoothly on the small OLED screen.

Each game program must define the ADC pin, initialise the 64 × 32 OLED display, and configure I²C communication for efficient data transfer.

For this system, sample codes for two popular games, Flappy Bird and Pong, have already been developed. These examples provide an excellent starting point and clearly demonstrate how simple yet engaging games can run effectively on minimal hardware.

Once the code is ready and downloaded from the source portal, select ATtiny85 as the target board in the Arduino IDE, choose the correct clock setting—typically an 8MHz internal clock—and use a compatible programmer, such as USBasp or an Arduino configured as an ISP, to program the code. After programming, the tiny Game Boy is ready to play. Fig. 3 shows a code snippet that configures the display, pixel rendering, and potentiometer control within the program.

```
gameBoy_Pong.ino
1  #include <Wire.h>
2  #include <SSD1315.h>
3
4  SSD1315 display;
5
6  #define W 64
7  #define H 32
8
9  #define POT_PIN 1
10
11 int playerY = 10;
12 int aiY = 10;
13
14 int ballX = 32;
15 int ballY = 16;
16 int vx = 1;
17 int vy = 1;
18
19 void resetBall() {
```

Fig. 3: Game Boy code snippet

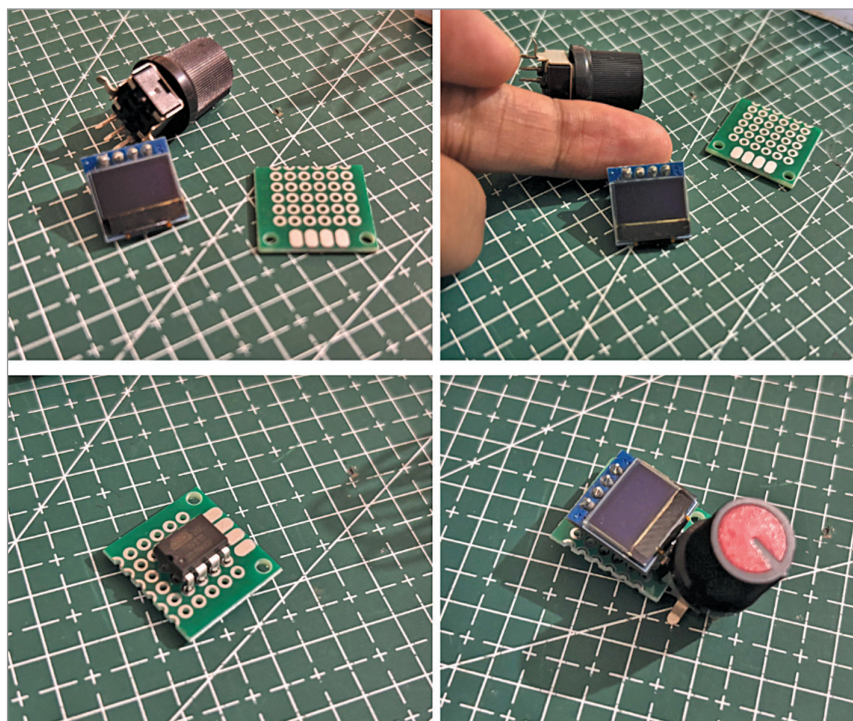


Fig. 4: Soldering of the Game Boy components